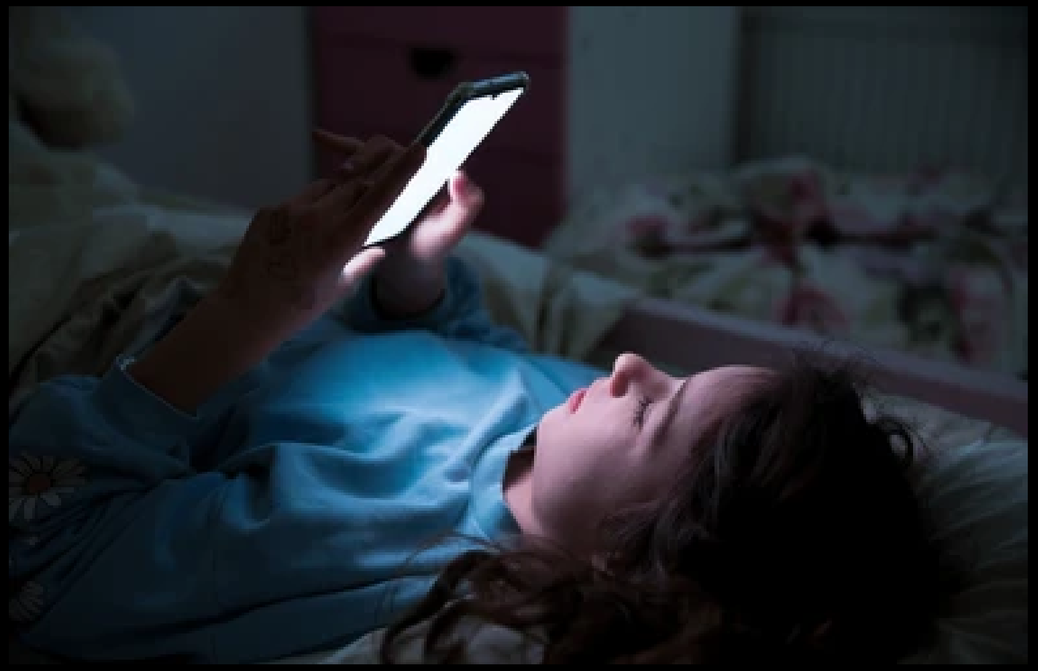
Digital Exposure & Health Impacts: Screen Time Analysis of Indian Kids (2025)



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Introduction

In the post-pandemic digital era, screen usage among children has skyrocketed due to online learning, entertainment, and social engagement. While screens provide benefits, extended exposure—especially for recreational purposes—can lead to health challenges like poor sleep, anxiety, and eye strain. This project analyzes the screen usage habits of Indian kids in 2025 based on gender, age, device preferences, content types, and location (urban/rural), to understand behavioral trends and their associated health outcomes.



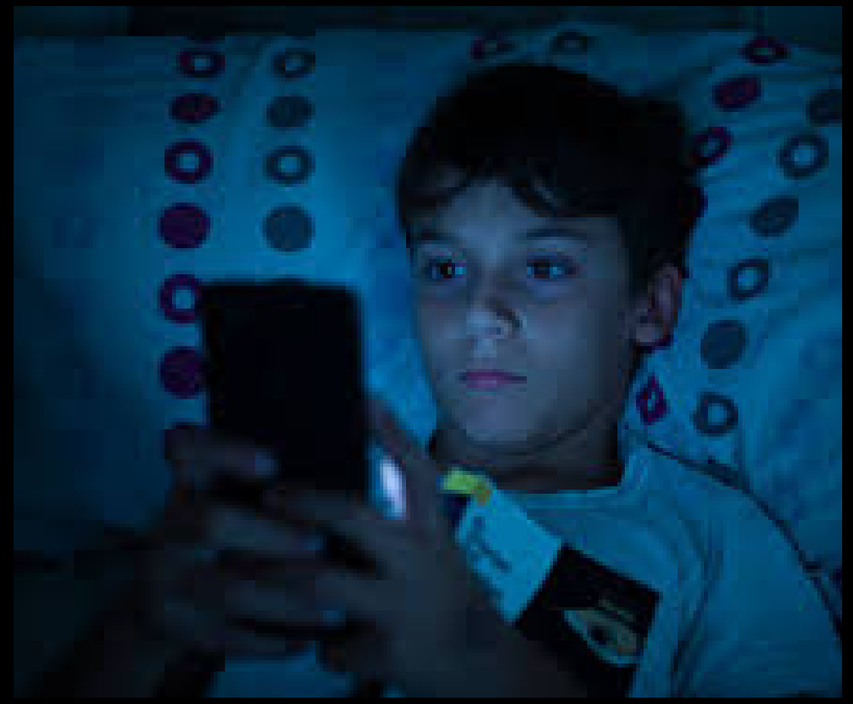
Problem Statement

The increased reliance on digital screens among Indian children, especially post-COVID, has blurred the line between healthy and excessive screen use. Many children are spending extended hours on recreational screen activities, often exceeding recommended limits. This raises concerns about long-term health outcomes, such as poor sleep, anxiety, and eye strain. Despite this, there is limited visibility into which age groups, genders, or regions are most at risk, and whether screen time is being used productively or not.



🎯 Objective

- Analyze screen usage patterns by age, gender, and location
- Compare educational vs. recreational screen habits
- Identify kids exceeding healthy screen time limits
- Detect health issues linked to screen overuse
- Provide data-driven recommendations for better digital health



🛠️ Tool Used

- **SQL**: For data extraction and analysis
- **Excel**: For data visualization
- **Canva**: For creating the presentation layout



Data Overview

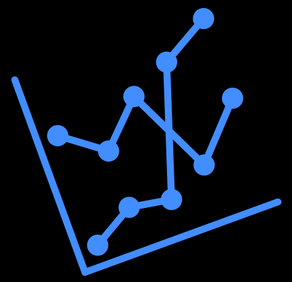


Dataset includes the following key columns:

Column	Description
Age	Age of the child (8–18)
Gender	Male/Female
Avg_Daily_Screen_Time_hr	Screen time in hours
Primary_Device	Most-used device (Phone, TV, Laptop, etc.)
Exceeded_Recommended_Limit	TRUE/FALSE — over 2 hours per day
Educational_to_Recreational_Ratio	Ratio of educational to entertainment use
Urban_or_Rural	Living environment
Health_Impacts	Reported health issues (e.g., Poor Sleep)

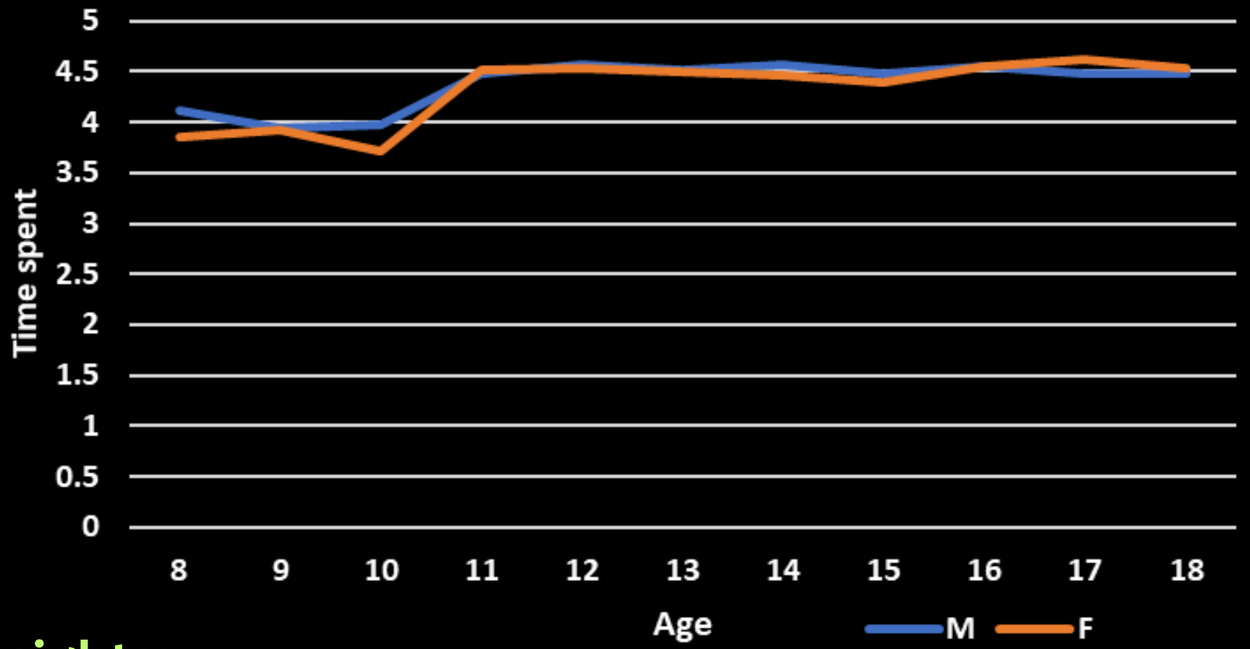
Quick Stats

- Total no. of records / rows 9712
- Total fields / Columns ---------- 8
- Male and Female Percentage 50.89% and 49.11% respectively
- Average Age ---------- 13
- Average Screen Time 4.35 Hours
- Percentage of kids exceeding the recommended limit –-- 85.47%





Average screen time by gender and age



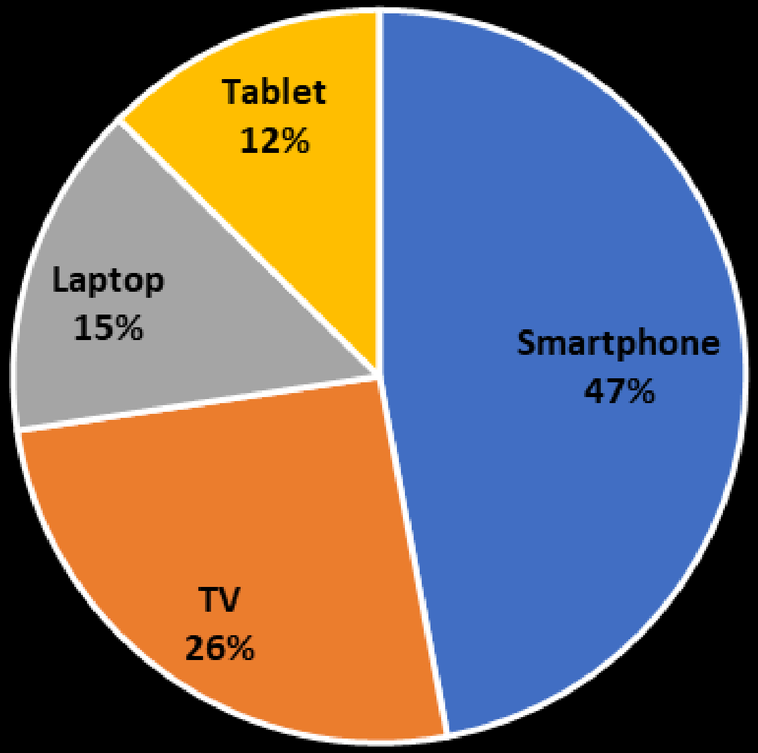
Insights:

- Teenagers (particularly ages 14–17) show the highest screen time, averaging over 5 hours/day in some age-gender segments.
- There is a steady increase in screen time from age 11 to 16, after which it slightly drops.
- This pattern suggests a behavioral shift around early-to-mid adolescence, possibly due to school pressure, social media use, or streaming habits.

Most Used Device

Insights :

- Smartphones are the dominant device, followed by Laptops and TVs.
- The convenience and portability of smartphones may explain their high usage
- Tablets and desktops are the least used, showing a strong trend toward mobile-first consumption.
- The heavy reliance on smartphones may be contributing to sleep disruption and constant stimulation, especially at night.



✅ **1. "Educational" screen content**

This means screen time used for learning or studying, such as:

- Watching online classes
- Reading e-books
- Using educational apps (e.g., Duolingo, Khan Academy)
- Doing schoolwork on a computer

✅ **2. "Recreational" screen content**

This means screen time used for fun or entertainment, such as:

- Watching YouTube, Netflix, or movies
- Playing video games
- Browsing social media
- Scrolling memes or funny videos

✅ **3. "Ratio"**

A ratio is a way to compare two numbers.

In this case:

👉 Educational Time ÷ Recreational Time

🧮 Example Calculations:

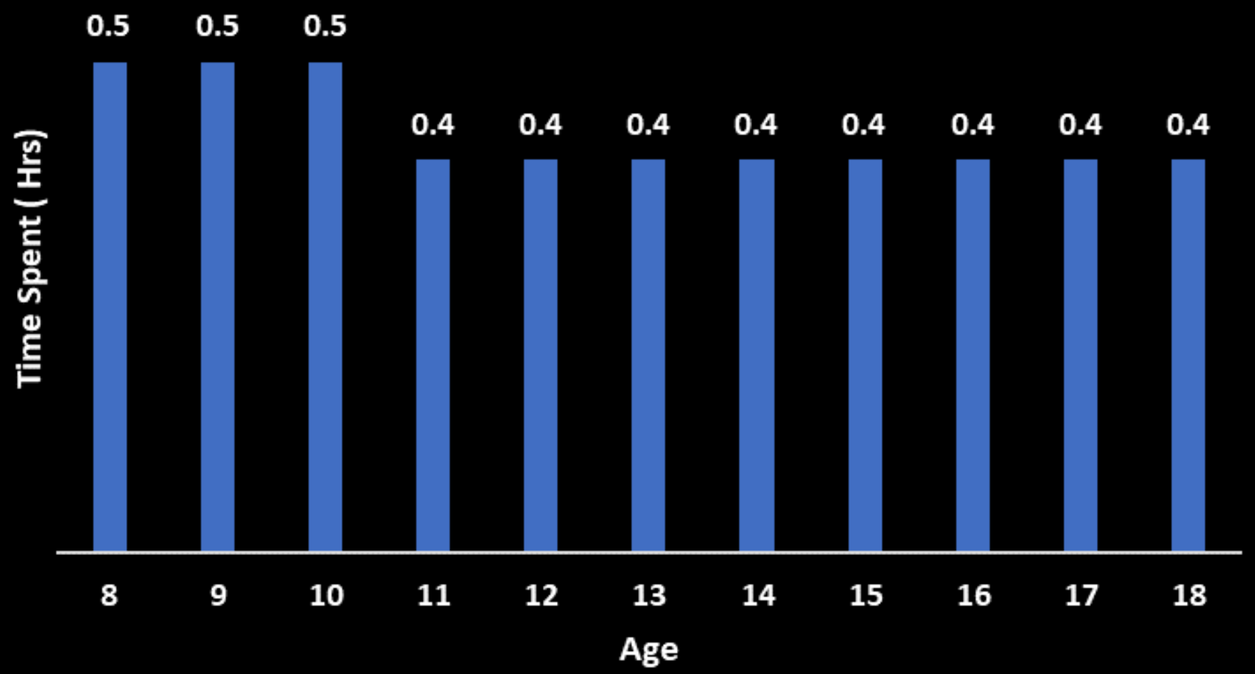
Educational Time ÷ Recreational Time

Educational Time	Recreational Time	Ratio (E ÷ R)	Meaning
30 mints	60 mints	0.5	Half the time is educational, the rest is fun
60 mints	60 mints	1	Equal time spent on education and fun
0 mints	60 mints	0.00	Only fun, no learning

Educational vs. Recreational Ratio by Age

Insights:

- The average educational-to-recreational ratio remains below 0.4 across all ages — meaning screen usage is largely entertainment-focused.
- Younger children (age 10–12) show a slightly better balance, while older teens use screens predominantly for recreation.
- This imbalance highlights a growing concern: digital consumption is more about passive entertainment than active learning.



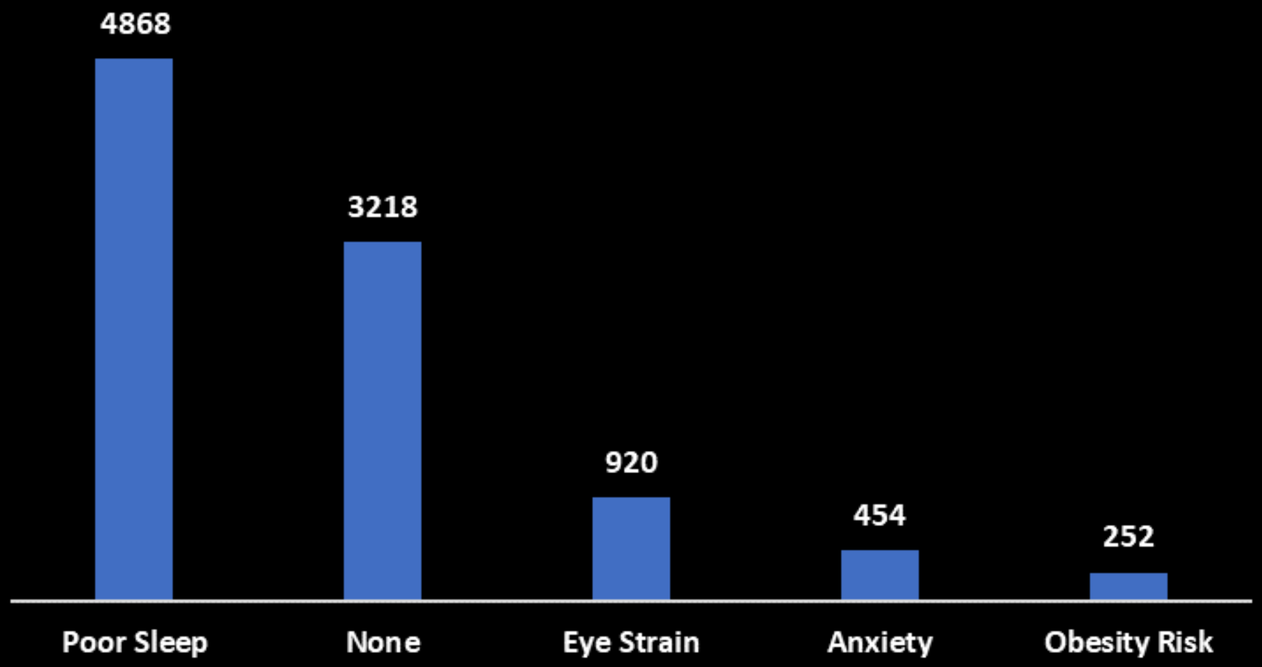
Urban vs Rural: Average Screen Time



Insights:

- Rural children spend more average screen time (~4.37 hrs/day) than urban children (~4.34 hrs/day)
- Fewer recreational alternatives (like parks, clubs) in rural regions
- Less digital supervision or awareness of screen health risks
- Higher dependence on screens for both learning and leisure in rural settings

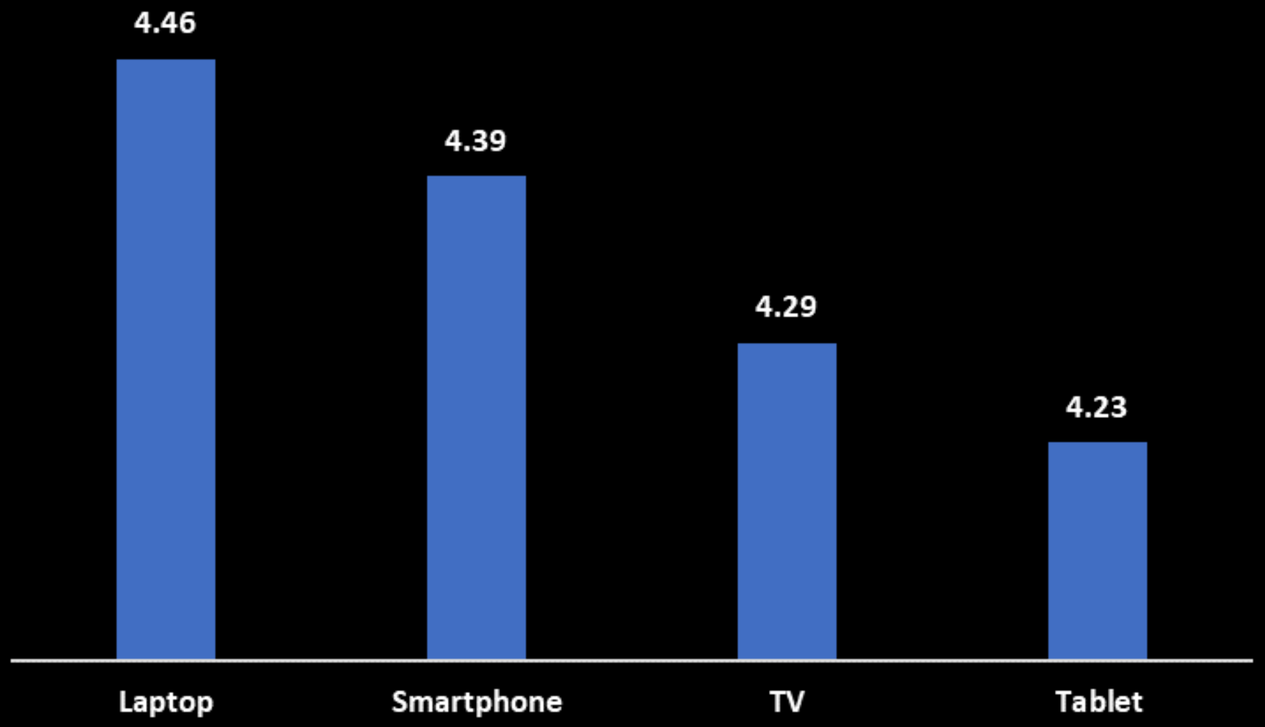
Health Impact



Insights:

- The most reported health issue is “Poor Sleep”, followed by “Eye Strain” and “Anxiety”.
- These effects are more commonly reported among teens aged 14–17, especially in rural areas and among heavy recreational users (Edu/Rec ratio < 0.3).

Device used and Avg. screen Time and Health



Insights:

- Kids who use Laptops and Smartphones report the highest average screen times, with many exceeding 5 hours/day.
- These users also show more frequent health issues, especially related to sleep and eye strain — pointing to both device type and screen duration as key risk factors.

Insights Highlight

✅ From the dataset, it’s clear that female children aged 14–16 from rural areas spend the highest average screen time, mostly on smartphones, with a very low educational-to-recreational ratio.

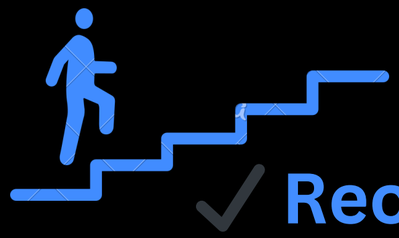
These kids are not only exceeding healthy usage limits but also report the most severe health symptoms, especially poor sleep and eye strain.



🔍 Overall Insights from the Project



1. Screen time is highest among teenagers aged 14–17, especially among girls in urban areas.
2. Over 85% of kids exceed the 2-hour daily screen time recommendation, raising serious concerns.
3. Smartphones are the most-used devices, suggesting constant and portable access to digital media.
4. The educational vs. recreational screen ratio is low (avg < 0.4), confirming that most screen time is spent on entertainment.
5. Rural children are more digitally exposed than urban ones, but this also puts them at higher risk of health impacts.
6. The most reported health issue is “Poor Sleep”, likely due to late-night screen use, followed by eye strain and anxiety.



Recommendations:

🧭 Introduce structured screen time plans in schools and households.
🌃 Restrict screen usage 1 hour before bedtime to improve sleep quality.
🎯 Promote educational gamified content to improve learning engagement.
📚 Conduct workshops for parents and children about the safe use of digital devices.
🧘 Encourage offline activities like sports, reading, or mindfulness to balance digital life.

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